

PRACTICAL 2

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
101	The Alchemist	9780061122415	1988
102	Harry Potter	9780747532743	1997
103	The Hobbit	9780261102217	1937
104	Atomic Habits	9780735211292	2018
105	1984	9780451524935	1949

```
5 rows in set (0.00 sec)
```



```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	1	101	2026-08-20
2	2	102	2026-08-21
3	3	104	2026-08-22

```
3 rows in set (0.00 sec)
```



```
mysql> SELECT
->     M.member_name,
->     B.title
-> FROM Members M
-> JOIN Loans L
->     ON M.member_id = L.member_id
-> JOIN Books B
->     ON L.book_id = B.book_id;
```

member_name	title
Varshini	The Alchemist
Rahul	Harry Potter
Priya	Atomic Habits

```
3 rows in set (0.00 sec)
```

? **JOIN** → Combines related tables.

? **GROUP BY** → Groups similar values.

? **COUNT()** → Counts records.

? **Transaction** → Treats multiple operations as one unit.

? **COMMIT** → Saves changes permanently.

? **ROLLBACK** → Cancels changes.

? **Index** → Speeds up searching.

```

mysql> SELECT
->     published_year,
->     COUNT(book_id) AS total_books
-> FROM Books
-> GROUP BY published_year;
+-----+-----+
| published_year | total_books |
+-----+-----+
|          1988 |           1 |
|          1997 |           1 |
|          1937 |           1 |
|          2018 |           1 |
|          1949 |           1 |
+-----+-----+
5 rows in set (0.00 sec)

mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY AUTO_INCREMENT,
->     book_id INT,
->     donation_date DATETIME DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> BEGIN;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books
-> (book_id, title, isbn, published_year)
-> VALUES
-> (106, 'Clean Code', '9780132350884', 2008);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History
-> (book_id)
-> VALUES
-> (106);
Query OK, 1 row affected (0.01 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

```

```
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```

book_id	title	isbn	published_year
101	The Alchemist	9780061122415	1988
102	Harry Potter	9780747532743	1997
103	The Hobbit	9780261102217	1937
104	Atomic Habits	9780735211292	2018
105	1984	9780451524935	1949
106	Clean Code	9780132350884	2008

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	donation_date
1	106	2026-08-22 22:42:19

```
1 row in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO Books
  -> (book_id, title, isbn, published_year)
  -> VALUES
  -> (107, 'Database Systems', '9781234567890', 2020);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
101	The Alchemist	9780061122415	1988
102	Harry Potter	9780747532743	1997
103	The Hobbit	9780261102217	1937
104	Atomic Habits	9780735211292	2018
105	1984	9780451524935	1949
106	Clean Code	9780132350884	2008
107	Database Systems	9781234567890	2020

```
7 rows in set (0.00 sec)
```

```
mysql> CREATE INDEX idx_isbn
  -> ON Books(isbn);
Query OK, 0 rows affected (0.00 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SHOW INDEX FROM Books;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
books	0	PRIMARY	1	book_id	A	7	NULL	NULL		BTREE			YES	NULL
books	1	idx_isbn	1	isbn	A	7	NULL	NULL	YES	BTREE			YES	NULL

```
2 rows in set (0.05 sec)
```

```
mysql> SELECT *
  -> FROM Books
  -> WHERE isbn = '9780132350884';
```

book_id	title	isbn	published_year
106	Clean Code	9780132350884	2008

```
1 row in set (0.00 sec)
```

```
mysql>
```